

Testing Microservices for Scalability and Fault Tolerance Under Unpredictable Loads

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This paper presents a systematic review on microservice testing, focusing on research published since 2023. The author is affiliated with FAST National University, Department of Software Engineering. Correspondence regarding this research can be directed to K248204@nu.edu.pk.

Abstract

The ascent of microservices engineering has reformed programming configuration by empowering versatility, adaptability, and quick turn of events. Notwithstanding, the dynamic and conveyed nature of microservices presents huge difficulties in testing, especially in reenacting genuine burden changeability and distinguishing disappointment focuses. Current testing structures miss the mark in giving far reaching answers for assessing the flexibility and execution of microservices under unusual traffic examples and disappointment conditions. This paper presents a clever testing structure intended to address these holes by consolidating progressed recreation methods for load changes and issue infusion in dispersed microservices conditions. The proposed structure works with early discovery of framework bottlenecks and disappointment focuses via robotizing stress testing situations intelligent of certifiable functional difficulties. Our methodology is approved through contextual analyses showing its viability in further developing the adaptation to non-critical failure and adaptability of microservices-based applications. The outcomes demonstrate the way that this system can essentially upgrade the unwavering quality of microservices structures, offering a viable answer for engineers and analyzers popular creation conditions.